Source: CSNI-DOC-SRD-JRB CHL ENFORCEMENT.pdf.

1. Authentication and Access

Use the existing CHL authentication and authorization services.

Submodule	Features
Login	Sign in with existing CHL credentials; email and password fields; password visibility control.
Password Recovery	Access the existing CHL forgot-password process.
Session Management	Maintain the authenticated session, refresh authentication through the existing service, log out, and handle session expiry.
Role-Based Access	Display only modules and records authorized for the signed-in administrator, supervisor, or Principal Officer.
Geographical Access	Limit accessible monitoring data according to centrally assigned geographical permissions.
Profile / Access Details	Proposed read-only view of the signed-in user's identity, organization, role, and assigned scope.

Boundary: User creation, role assignment, account deactivation, and permission configuration remain in the existing CHL system.

The document specifies a 24-hour session for the mobile application; whether the dashboard uses the same duration needs confirmation.

2. Overview / Monitoring and Corporate Dashboard

This is the high-level view of enforcement operations.

2.1 Summary Metrics

The document explicitly supports:

- Total agent verifications.
- Active-trip verification results.
- Inactive/no-active-trip verification results.

- Violations by agent.
- Unresolved violations.
- Repeat vehicles without appropriate action.
- Agent activity by location.

These should respect the user's authorized geographical scope.

Important: "Active-trip verification results" counts verification outcomes; it should not automatically be labeled as the number of unique active trips.

2.2 Enforcement Performance

- View verification activity across agents.
- View violations issued by each agent.
- Monitor unresolved cases.
- Identify repeated reports without follow-up.
- Review enforcement activity by location.

2.3 Recent Activity

A proposed consolidated view of the activity records described in the document:

- Vehicle plate number.
- Verification method.
- Verification result.
- Agent.
- Date and time.
- Action status.
- Enforcement location.

2.4 Corporate Monitoring View

The document names a **Monitoring and Corporate Dashboard**, but does not define separate corporate KPIs.

For the initial scope, this can summarize the documented enforcement metrics across permitted locations. Revenue targets, financial forecasts, executive scorecards, and other corporate KPIs would require additional requirements.

3. Command and Control Centre

A centralized operational view bringing together enforcement activity, reported incidents, and cases requiring attention.

The document explicitly names this dashboard, but its detailed layout and response procedures are not specified. The following submodules are proposed groupings of documented information.

Submodule	Features
Operational Activity Feed	Incoming verification records, violations, payment updates, and synchronized field activity received by CHL.
Cases Requiring Attention	Unresolved violations, escalated violations, and repeat vehicles without follow-up.
SOS Monitoring	Display SOS notifications submitted by agents, where exposed by the backend.
Field Reports	Display vehicle reports, driver reports, situation reports, and general notifications submitted from the mobile application.
Loaded / Empty Reports	Display agents' reported haulage-loaded or empty status where recorded.
Incident Context	Show related agent, vehicle, location, timestamp, case, and evidence where supplied.
Location-Based Operational View	Show where recorded verification and enforcement events occurred. A map is a proposed presentation option.

Not established by the document: Emergency dispatch, two-way chat, remote commands, CCTV, continuous vehicle tracking, or continuous agent tracking.

GPS captured during an enforcement event must not be represented as a continuously updated live location.

4. Geographical Monitoring

This provides the geographical views you identified.

Submodule	Monitoring scope
Enforcement Location	Activity associated with a specific enforcement point/location.
Enforcement Zone	Activity within a centrally defined enforcement zone.
Ward	Activity associated with a ward.

Local Government Area	Activity within an LGA.
State	Activity across a state.
Region	Activity within a centrally defined region.
Subregion	Activity within a centrally defined subregion.
National	Consolidated national visibility for authorized users.

Features available within each scope

- Verification totals and outcomes.
- Agent activity.
- Violations issued.
- Unresolved and escalated cases.
- Repeat-vehicle alerts.
- Related payment/compliance status.
- Enforcement locations and recorded GPS positions.
- Access to underlying records.

These are **different views of the same underlying data**, not eight separate systems.

The parent-child relationships between zones, locations, regions, and subregions must come from CHL configuration rather than being assumed.

5. Agent and Organization Monitoring

This module monitors existing agents; it does not create or administer their accounts.

Submodule	Features
Agent Identity	Agent ID, name, and vendor/organization where available.
Assigned Scope	View centrally defined geographical/enforcement scope where exposed by CHL.
Verification Activity	View each agent's verification records and outcomes.
Enforcement Activity	View violations issued, evidence submitted, demand notices, and action status.
Activity by Location	View recorded enforcement points and GPS-associated activity.

Follow-Up Monitoring	Identify unresolved cases and repeated vehicle reports involving the agent.
Performance Overview	Review documented verification and enforcement metrics by agent.

Not part of the current document: Automated agent scoring, rankings, attendance, shift scheduling, or payroll. Agent performance scoring is listed as a future extension.

6. Vehicle Verification and E-Tag Monitoring

The dashboard displays verification results produced through the mobile application and CHL services. It does not need to operate scanning hardware itself.

6.1 Verification Records

- RFID verification records.
- QR verification records.
- Manual plate verification records.
- OCR-confirmed plate verification records.
- Successful and unsuccessful attempts.
- Live-backend versus cached/offline verification indicators.
- Associated agent, timestamp, location, and action.

6.2 Vehicle Identity

- Vehicle ID where resolved.
- Plate number.
- Vehicle type where available.
- Related verification, trip, and violation records.

6.3 E-Tag Results

- Valid E-Tag.
- Missing E-Tag.
- Invalid/fake E-Tag.
- Tag-vehicle mismatch.

6.4 Verification Outcomes

- Active Trip.
- No Active Trip.
- E-Tag Missing.

- Invalid E-Tag.
- Tag-Vehicle Mismatch.
- Vehicle Not Found.
- No Trip History.
- Verification Unavailable.
- Expired/Previous Trip.

6.5 Verification Detail

- Verification method and outcome.
- Related vehicle and trip.
- Action taken.
- Violation issued, if any.
- Payment request generated, if any.
- Pending action.
- Synchronization status.

Rules to preserve: A network failure is not “No Active Trip,” and a failed RFID scan is not proof of a missing tag. Raw RFID values should not appear in general operational views; any privileged audit-identifier exposure needs explicit authorization.

7. CHL Trip Monitoring

A read-only view of trip information returned by the existing CHL system.

Submodule	Features
Active Trip Details	View the backend-authoritative current trip and its status.
No Active Trip	View verified vehicles for which CHL returned no active trip.
Most Recent Previous Trip	View the most recent trip when no active trip exists.
No Trip History	Clearly identify vehicles with no recorded CHL trips.
Destination Detection	View backend-reported destination detection/completion information where available.
Post-Payment Trip Status	Observe trip activation or updated compliance status following confirmed payment.

Trip detail fields

- Transaction reference.

- Trip start date/time.
- Payment date/time.
- Vehicle plate number.
- Vehicle type.
- Transaction type.
- Originating state and LGA.
- Destination state and LGA.
- Trip status.
- Trip validity countdown.

Boundary: The dashboard must not calculate its own trip validity, expire trips merely because a countdown reaches zero, or create trips independently.

8. Violation and Case Monitoring

This is a monitoring module, not a case-editing module under your read-only scope.

8.1 Violation Register

- View violations received by CHL.
- Open individual case/violation records.
- View the associated vehicle, agent, location, time, evidence, and notes.
- View linked demand notices and payment status.

8.2 Violation Categories

The initial categories specified are:

- No Active CHL Trip.
- No E-Tag.
- Fake/Invalid E-Tag.
- Tampered E-Tag.
- Expired Trip.
- Refusal to Comply.

Tag-vehicle mismatch must also be visible as a documented breach, although the SRD does not explicitly include it in the initial category list.

8.3 Status Views

- Payment Pending.
- Resolved.
- Escalated.
- Cancelled.

- Any additional internal states returned by the backend.

8.4 Unresolved Cases

- View outstanding violations.
- View recorded actions and pending follow-up.
- Identify related repeat-vehicle alerts.

8.5 Closure and Cancellation History

- View whether a case was closed or cancelled.
- View the responsible actor and associated audit information where available.

Scope distinction: The SRD allows authorized administrators and other roles to close or cancel violations. Your clarification makes this dashboard read-only, so those action buttons are excluded unless you explicitly expand its scope.

9. Evidence and Enforcement Records

A read-only viewer for evidence captured through the mobile application.

Submodule	Features
Photograph Viewer	View multiple photographs associated with a violation.
Vehicle Evidence	Front, rear, side, and haulage-bay photographs.
E-Tag Evidence	Missing-tag, fake/invalid-tag, and other relevant photographs.
Evidence Metadata	Capture timestamp, GPS coordinates/location, agent identity, and violation reference where available.
Enforcement Notes	Read agents' submitted notes.
Related Records	Access the associated vehicle, verification, violation, and enforcement location.

The dashboard should not upload replacement evidence, edit photographs, or alter captured metadata.

The document requires evidence capture through the mobile camera and prohibits gallery uploads during enforcement evidence collection.

10. Demand Notice, Payment and Compliance Monitoring

Submodule	Features
Demand Notice Register	View generated notices/invoices and their associated violations.
Notice Details	Vehicle information, CHL levy, penalty, total payable, and payment reference.
Payment Method Details	Display the backend-provided method: QR, USSD, transfer, generated account/reference, POS/card, or existing CHL platform.
Payment Status	Monitor Payment Pending and Payment Received, as reported by CHL.
Post-Payment Compliance	Observe trip activation, tagging-process initiation, or the applicable compliance outcome.
Related Transactions	View associated transaction references and payment timestamps.

Boundary: No tariff editing, invoice generation, manual payment confirmation, refunds, or independent reconciliation controls are included in the read-only dashboard.

“Payment received” and “trip activated” should remain distinct statuses.

11. Notifications and Repeat-Vehicle Alerts

11.1 Repeat-Vehicle Alerts

- Identify a vehicle reported **three qualifying times without appropriate follow-up**.
- View the related reports.
- View involved agents and locations.
- View associated cases and recorded actions.
- Display notifications supplied by the central supervisory process.

This is not an alert after any three scans.

11.2 Enforcement Notifications

- Repeat-vehicle notifications.
- Payment/compliance notifications where supplied.
- Other backend-provided enforcement notifications.

11.3 Field Notifications

- SOS.
- Vehicle reports.
- Driver reports.
- Situation reports.
- General system notifications from agents.
- Loaded/empty confirmations.

11.4 Operational Notifications

The SRD lists:

- RFID reader connected/disconnected.
- Internet disconnected/restored.
- GPS disabled.
- Pending offline actions.
- Successful synchronization.
- App update available.
- Session expiring.

These are explicitly application notifications. Displaying remote agents' operational notifications on the Admin Dashboard requires the backend to receive and expose them; the SRD does not guarantee that telemetry.

12. Verification History and Audit Trail

12.1 Verification History

- Browse recorded verification attempts.
- View completion status.
- View actions taken.
- View linked violations.
- View payment requests generated.
- View pending actions.
- Open record details.

The SRD's "last 30 days" default and "view all" behavior apply to personal agent history. They should not be treated as a 30-day limit on administrator access.

12.2 Complete Audit Trail

The document explicitly gives authorized System Administrators access to the complete audit trail.

Audit fields include:

- Agent ID and name.
- Vendor/organization where applicable.
- Date and time.
- Verification method.
- Processed identifier, subject to access rules.
- Vehicle ID and plate were resolved.
- Verification result.
- Trip reference.
- Violation reference.
- GPS location.
- Enforcement point.
- Device/application metadata where available.
- Action taken.
- Synchronization status.

12.3 Record Relationships

A proposed navigation feature connecting existing records:

- Verification of vehicle/trip.
- Verification to violation.
- Violation to evidence.
- Violation to demand notice/payment.
- Record to agent/location.

12.4 Authorized Visibility

- Complete audit access for authorized administrators.
- Restricted access for supervisors according to backend permissions.
- No editing or deleting audit records.

13. Synchronization and Field Operational Status

This module distinguishes received data from field information that may be delayed.

Submodule	Features
Record Synchronization Status	Display the synchronization status stored with received records.
Offline-Origin Activity	Identify records created while offline, where the backend supplies this information.

Evidence Synchronization	Show available evidence/upload status where exposed.
Agent Device Status	Display reported RFID, internet, and GPS status only if supplied by CHL.
Data Freshness	Proposed timestamps showing when status was reported or data was last refreshed.

Critical limitation: The dashboard cannot know about actions still stored exclusively on an offline phone. It must not show a last-reported device state as guaranteed current status.

A centralized device-health dashboard and pending-upload counts are not explicitly required by the SRD.

Shared dashboard controls

These are **proposed interface features**, needed to make the documented monitoring data usable, rather than explicitly specified dashboard requirements:

- Geographical scope selector.
- Date/time-range filter.
- Agent and organization filters.
- Vehicle plate/reference lookup.
- Verification-method and result filters.
- Violation-category and case-status filters.
- Payment-status filter.
- Record-detail navigation.
- Clear loading, empty, error, and last-updated states.

These controls filter or display data; they do not change CHL business records.

Features not to add to the current scope

Do not duplicate existing CHL administration:

- Account, vendor, or Principal Officer creation.
- Vehicle registration.
- Tag issuance.
- Tariff configuration.
- Trip-rule configuration.
- Role or geographical-assignment editing.

Also keep the SRD's future extensions separate from the initial dashboard:

- Automated vehicle classification.
- AI-assisted haulage detection.
- Driver identification.
- Digital impound workflows.
- Advanced supervisor case management.
- Advanced geofencing.
- Automated fraud detection.
- Agent performance scoring.
- Advanced national enforcement analytics.

In summary: the dashboard consists of access, overview, Command Centre, geographical monitoring, agent monitoring, verification/E-Tag monitoring, trip monitoring, violation monitoring, evidence viewing, payment monitoring, notifications, audit/history, and synchronization visibility all powered by the existing CHL system.